1. Identity

Document	STAR-PKG-STAR-SUN-EN · v2.0 · 2026-10-21
Package	star-sun
Version	0.1.0
Capability	star.orbit.sun
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_sun

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_sun-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_sun-0.1.0-py3-none-any.whl	efe79a6c0bc738b3ecf83c90e2e34c67a53b7d6eff223b175c5da48d5d45942e
star_sun.py	9c1a0f1b89694b7cadd6d21fc0c13a2af1ef4533a24740c8eda6fc9fe461aaa7
star_sun-0.1.0-source.zip	9c7264b8b2753975aeebc0e416f9c45f82954528174f8e32f9301de1c1947e72

5. What it does, and its limits

Low-precision geocentric Sun vector for 1950-2050 (measured within 0.0097 deg of JPL DE440 apparent), orbit beta angle and cylindrical shadow test; every function returns finite values or raises ValueError.

Not an ephemeris: 0.01 deg is about 36 arcseconds. No nutation (mean equator of date, about 0.7 deg from J2000 in 2050: rotate before mixing with J2000/GCRF vectors). The shadow is a cylinder: no penumbra, no oblateness, no atmosphere; near the shadow edge the answer can differ from a conical model by tens of seconds of orbit.

- `sun_vector_au(jd_day, jd_frac=0.0)` and `sun_ra_dec_deg`: geocentric Sun in AU, mean equator and equinox of date, for UT1 Julian dates from 1950-01-01 to 2050-01-01 (inclusive); outside, `ValueError`. Reproduces Vallado Example 5-1 within 1.2e-6 AU.
- Measured accuracy (402 dates): direction within 0.0097 deg of JPL DE440 *apparent* and of astropy, distance within 7e-5 AU. The series includes aberration: against *geometric* positions the difference reaches 0.014 deg.
- `beta_angle_deg(sun, inc_deg, raan_deg)`: angle between the Sun direction and the orbit plane, in [-90, 90], positive on the side of the angular momentum; only the direction of `sun` is used.
- `in_shadow(r_sat, sun, body_radius=6378.137)`: True when the satellite is behind the body and closer than one body radius to the Sun-body axis (cylinder, parallel rays, no penumbra); `r_sat` and `body_radius` in the same unit.
- Every function returns finite values or raises `ValueError`: non-numeric, boolean, NaN, infinite or |value| >= 1e300 input; a vector that is not three numbers; a zero Sun vector; a satellite inside the body; a non-positive body radius; inclination outside [0, 180]; node outside [-360, 360]; a date outside 1950-2050.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_sun_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-sun}: Sun direction, beta angle and eclipse test},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_sun/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts